

Niklas Rieken

Curriculum Vitae

Geschwister-Scholl-Straße 26

44135 Dortmund, Germany

+49 1523 8736069

rieken@oms.rwth-aachen.de

tnick.github.io

in niklas-rieken

Personal Data

Date of Birth November 9, 1992

Place of Birth Krefeld, Germany

Education

- November 2025 **Doctorate (PhD)**, *RWTH Aachen University*, Economics
Thesis: Matroid Optimization in Auction Theory
Final Grade: summa cum laude (highest distinction)
- May 2019 **Master of Science**, *RWTH Aachen University*, Computer Science
Thesis: Network Pricing Games with Atomic Splittable Followers
Final Grade: 1.4
- March 2017 **Bachelor of Science**, *RWTH Aachen University*, Computer Science
Thesis: Alphabets and Machines as Structures and Their Decision Problems
Final Grade: 2.5
- May 2012 **Abitur (German A-Levels)**, *Michael-Ende-Gymnasium*, Tönisvorst
Advanced Courses: Math and Physics
Final Grade: 2.6

Work Experience

- 2019–2025 **Doctoral Researcher**, *Chair of Management Science, RWTH Aachen University*
Researcher in Combinatorial Optimization and Algorithmic Game Theory.
Matchings, matroids, mechanism design, teaching courses in mathematical economics.
- 2017–2019 **Research Assistant**, *Chair of Operations Research, RWTH Aachen University*
Developer for GCG (*generic column generation*, MIP solver).
Performance tests, extracting statistics, and collecting information on vast instance sets.
- 2014–2019 **Teaching Assistant**, *Chair of Computer Science 7 / TCS, RWTH Aachen University*
TA for undergraduate courses in theoretical computer science.
Teaching exercise groups, grading assignments and exams.

Scientific Publications

- OR Letters 2025 **A Simplified Analysis of the Ascending Auction to Sell a Matroid Base**
with Britta Peis
- ANRW 2025 **Using Explicit (Host-to-Host) Flow Measurements for Network Tomography**
with Ike Kunze, Constantin Sander, Alexander Ruhrmann, and Klaus Wehrle
- TEAC 2025 **Faster Dynamic Auctions via Polymatroid Sum**
with Katharina Eickhoff, Meike Neuwohner, Britta Peis, Laura Vargas Koch, and László A. Végh
- Networks 2024 **A flow-based ascending auction to compute buyer-optimal Walrasian prices**
with Katharina Eickhoff, S. Thomas McCormick, Britta Peis, and Laura Vargas Koch
- WINE 2023 **Faster Ascending Auctions via Polymatroid Sum**
with Katharina Eickhoff, Britta Peis, Laura Vargas Koch, and László A. Végh
- MAPSP 2022 **A Primal-Dual and Primal-Greedy Approximation Framework for Weighted Covering Problems**
with Britta Peis, José Verschae, and Andreas Wierz

Scientific Talks

July 3, 2025	A Self-Reflecting Greedy Algorithm for Submodular Cover 10th Gerhard Woeginger Research Colloquium, RWTH Aachen University
May 27, 2025	Combinatorial Optimization in Auction Design Seminar on Microeconomics, RWTH Aachen University
November 28, 2023	Selling Bases of a Matroid ORM PhD Seminar, RWTH Aachen University
August 30, 2023	An Ascending Auction for Computing Minimal Walrasian prices via the Matroid Partitioning Algorithm OR 2023, Universität Hamburg
June 15, 2023	Selling Bases of a Matroid 13th Day on Computational Game Theory, Vrije Universiteit Amsterdam
January 18, 2023	Computing Walrasian Prices via Matroid Partitioning Santiago Workshop on Combinatorial Optimization, Universidad de Chile
July 6, 2022	Computing Buyer-Optimal Walrasian Prices in Multi-Unit Matching Markets via a Sequence of Max-Flow Computations Seminario AGCO, Universidad de Chile

Awards

November 9, 2024	Prize for outstanding teaching activities Faculty 8 of Economics, RWTH Aachen University
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Coursework

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| ○ Convex Optimization | ○ Mathematical Logic I + II |
| ○ Operations Research I + II | ○ Logic and Games |
| ○ Optimization with Modelling Languages | ○ Applied Automata Theory |
| ○ Methods and Applications of Optimization | ○ Infinite Computations |
| ○ Discrete Differential Geometry | ○ Complexity Theory |
| ○ Satisfiability Checking | ○ Recursion Theory |

Skills

Operating Systems	GNU/Linux
Programming	Python, Bash, C/C++
Applications	L^AT_EX(+TikZ), Gurobi, git

Languages

German	native
English	proficient
Spanish	basics

Professional Interests

- Mathematical Optimization
- Theory of Formal Languages
- Algorithmic Game Theory

Personal Interests

- Table Tennis and Chess
- Guitar and Ukulele
- Reading, Movies, and Music
- Cooking, Board games